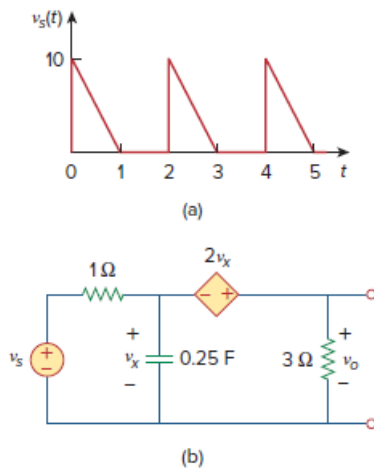


Problem

The signal in Fig. 17.76(a) is applied to the circuit in Fig. 17.76(b). Find $v_o(t)$.

Figure 17.76:



Step-by-step solution

Step 1 of 11

Refer to Equation 17.3 in the equation.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} -10(t-1), & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

Step 2 of 11

The time period is, $T = 2$.

Determine the value of ω_0 .

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \pi\end{aligned}$$

Determine the value of a_0 .

$$\begin{aligned}a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{2} \int_0^2 f(t) dt \\ &= \frac{1}{2} \int_0^1 f(t) dt + \frac{1}{2} \int_1^2 f(t) dt \\ &= \frac{1}{2} \int_0^1 -10(t-1) dt + \frac{1}{2} \int_1^2 0 \cdot dt\end{aligned}$$

Thus,

$$\begin{aligned}a_0 &= \frac{1}{2} \int_0^1 (10-10t) dt + 0 \\ &= \frac{10}{2} \left[t - \frac{t^2}{2} \right]_{t=0}^{t=1} \\ &= 5 \left[(1-0) - \left(\frac{1}{2} - 0 \right) \right] \\ &= \frac{5}{2}\end{aligned}$$

Determine the value of a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{2} \int_0^2 f(t) \cos(n\omega_0 t) dt \\
 &= \int_0^1 f(t) \cos(n\omega_0 t) dt + \int_1^2 f(t) \cos(n\omega_0 t) dt \\
 &= \int_0^1 [-10(t-1)] \cos(n\pi t) dt + \int_1^2 0 \cdot \cos(n\pi t) dt
 \end{aligned}$$

Thus,

$$\begin{aligned}
 a_n &= \int_0^1 -10(t-1) \cos(n\pi t) dt \\
 &= -10(t-1) \int_0^1 \cos(n\pi t) dt - \int_0^1 \frac{d}{dt}(-10(t-1)) \int \cos n\pi t dt \\
 &= -\frac{10}{n\pi} [(t-1) \sin(n\pi t)]_0^1 - \frac{10}{n^2 \pi^2} [\cos(n\pi t)]_0^1 \\
 &= 0 + \frac{10}{n^2 \pi^2} (1 - \cos(n\pi)) + 0 \\
 &= \frac{10}{n^2 \pi^2} (1 - \cos(n\pi))
 \end{aligned}$$

Thus,

$$a_n = \begin{cases} 0, & n \text{ is even} \\ \frac{20}{n^2 \pi^2}, & n \text{ is odd} \end{cases}$$

$$a_n = \frac{20}{(2n-1)^2 \pi^2}$$

Step 4 of 11

Calculate the coefficient, b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin(n\pi t) dt \\
 &= \frac{2}{2} \int_0^2 (-10(t-1)) \sin(n\pi t) dt \\
 &= \int_0^1 -10(t-1) \sin(n\pi t) dt + \int_1^2 (0) \sin(n\pi t) dt \\
 &= \int_0^1 -10(t-1) \sin(n\pi t) dt
 \end{aligned}$$

Solve the integral to calculate b_n .

$$\begin{aligned}
 b_n &= \int_0^1 -10(t-1) \sin(n\pi t) dt \\
 &= -10(t-1) \int_0^1 \sin(n\pi t) dt - \int_0^1 \frac{d}{dt}(-10(t-1)) \int \sin n\pi t dt \\
 &= \frac{10}{n\pi} [(t-1) \cos(n\pi t)]_0^1 - \frac{10}{n^2 \pi^2} [\sin(n\pi t)]_0^1 \\
 &= \frac{10}{n\pi} [0 - (-1)(1)] - \frac{10}{n^2 \pi^2} [\sin n\pi - 0]
 \end{aligned}$$

Since $\sin n\pi = 0$ for all n .

Thus,

$$b_n = \frac{10}{n\pi}$$

Step 5 of 11

The Fourier series is,

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= \frac{5}{2} + \sum_{n=1}^{\infty} \left\{ \left[\frac{20}{(2n-1)^2 \pi^2} \right] \cos \left(\frac{n\pi}{2} t \right) \right\} + \sum_{n=1}^{\infty} \frac{10}{n\pi} \sin n\pi t \\
 &= \frac{5}{2} + \frac{20}{\pi^2} \sum_{n=1}^{\infty} \left\{ \frac{1}{(2n-1)^2} \cos(n\pi t) \right\} + \frac{10}{n\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi t
 \end{aligned}$$

Or

$$v_s = \frac{5}{2} + \sum A_n \cos(n\pi t + \phi_n)$$

Where,

$$\begin{aligned}
 A_n &= \sqrt{\left(\frac{20}{(2n-1)^2 \pi^2} \right)^2 + \left(\frac{10}{n\pi} \right)^2} \\
 &= \sqrt{\left(\frac{400}{(2n-1)^4 \pi^4} \right) + \left(\frac{100}{n^2 \pi^2} \right)}
 \end{aligned}$$

And

$$\begin{aligned}
 \phi_n &= \tan^{-1} \left(\frac{\frac{10}{n\pi}}{\frac{20}{(2n-1)^2 \pi^2}} \right) \\
 &= \tan^{-1} \left(\frac{\pi(2n-1)^2}{2n} \right)
 \end{aligned}$$

Step 6 of 11

For the dc value of $\frac{5}{2}$, the capacitor behaves as open circuit.

Assume the current in this case be i .

Write the Kirchhoff's voltage law equation when only dc term is considered.

$$\begin{aligned}
 \frac{5}{2} + 2V_x - 4i &= 0 \\
 \dots\dots (1) \\
 2V_x - 4i &= -\frac{5}{2}
 \end{aligned}$$

Also, the voltage, V_x is,

$$\begin{aligned}
 V_x &= v_s - 1(i) \\
 V_x &= \frac{5}{2} - i \\
 V_x + i &= \frac{5}{2} \dots\dots (2)
 \end{aligned}$$

Solve (1) and (2).

$$V_x = 1.25 \text{ V and } i = 1.25 \text{ A}.$$

Step 7 of 11

Calculate the voltage, v_{o1} due to the dc term.

$$\begin{aligned}v_{o1} &= 3i \\&= (3)(1.25) \\&= 3.75 \text{ V}\end{aligned}$$

Therefore, the output voltage due to the dc term only is 3.75 V.

Step 8 of 11

Consider the ac term of the source voltage only.

For n^{th} harmonic,

$$\omega_n = n\pi,$$

$$\begin{aligned}X_{cn} &= \frac{1}{jn\pi(0.25)} \\&= -\frac{j4}{n\pi}\end{aligned}$$

Step 9 of 11

Apply KCL at the supernode.

$$\begin{aligned}\frac{v_s - V_x}{1} &= -\left[\frac{n\pi}{j4}\right]V_x + \frac{v_{o2}}{3} \\v_s &= \left(1 + \frac{jn\pi}{4}\right)V_x + \frac{v_{o2}}{3} \dots\dots (3)\end{aligned}$$

Step 10 of 11

Write the expression for the capacitor voltage in the circuit.

$$\begin{aligned}V_x &= v_{o2} - 2V_x \\V_x &= \frac{v_{o2}}{3}\end{aligned}$$

Substitute $\frac{v_{o2}}{3}$ for V_x in (3).

$$\begin{aligned}v_s &= \left(1 + \frac{jn\pi}{4}\right)\frac{v_{o2}}{3} + \frac{v_{o2}}{3} \\v_s &= \frac{v_{o2}}{3}\left(2 + \frac{jn\pi}{4}\right) \\v_s &= \frac{v_{o2}}{12}(8 + jn\pi) \\v_{o2} &= \frac{12v_s}{8 + jn\pi}\end{aligned}$$

Step 11 of 11

Using superposition.

$$\begin{aligned}
 v_o &= v_{o1} + v_{o2} \\
 &= 3.75 + \left(\frac{12}{8 + jn\pi} \right) v_s \\
 &= 3.75 + \left(\frac{12}{8 + jn\pi} \right) \left(\sum A_n \cos(n\pi t - \phi_n) \right) \\
 &= 3.75 + \sum V_n \cos(n\pi t - \phi_{v_o})
 \end{aligned}$$

Where

$$\begin{aligned}
 V_n &= \frac{12}{\sqrt{64 + n^2 \pi^2}} A_n \\
 &= \left(\frac{12}{\sqrt{64 + n^2 \pi^2}} \right) \left(\sqrt{\left(\frac{400}{(2n-1)^4 \pi^4} \right) + \left(\frac{100}{n^2 \pi^2} \right)} \right) \\
 \phi_{v_o} &= \tan^{-1} \left(\frac{n\pi}{8} \right) - \tan^{-1} \left(\frac{(2n-1)^2 \pi}{2n} \right)
 \end{aligned}$$

Therefore, the output voltage $v_o(t)$ is,

$$v_o(t) = 3.75 + \sum_{n=1}^{\infty} \left(\frac{12}{\sqrt{64 + n^2 \pi^2}} \right) \left(\sqrt{\left(\frac{400}{(2n-1)^4 \pi^4} \right) + \left(\frac{100}{n^2 \pi^2} \right)} \right) \cos \left(n\pi t + \tan^{-1} \left(\frac{n\pi}{8} \right) - \tan^{-1} \left(\frac{(2n-1)^2 \pi}{2n} \right) \right) V.$$